

Ashiqur Rahman Rahul

AI Systems Engineer | Embedded Systems | GPU & Edge Computing

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PROFESSIONAL SUMMARY

AI Systems Engineer with experience in AI product development, machine learning, embedded systems, GPU computing, and infrastructure-aware optimization. Currently working with the Cambridge Centre for Alternative Finance, University of Cambridge, on AI energy profiling, benchmarking, optimization strategy, and infrastructure-focused research. Interested in AI model development, AI systems engineering, embedded systems, machine learning, edge AI, intelligent infrastructure, and scalable deployment optimization.

CORE AREAS

AI Systems Engineering | AI Infrastructure & Optimization | Embedded Systems | GPU Computing & CUDA | Edge AI | Machine Learning & Computer Vision | LLM & RAG Systems | Applied Machine Learning & Deep Learning | Energy-Efficient Computing | AI Benchmarking & Analytics | IoT & Hardware Integration | Technical Research

PROFESSIONAL EXPERIENCE

AI Research Analyst - *Cambridge Centre for Alternative Finance (CCAF), University of Cambridge*

Remote, Berlin | 07/2025 - Present

- Lead research and product-oriented development on AI energy efficiency, infrastructure analysis, AI optimization, and intelligent deployment systems.
- Develop benchmarking and analytical frameworks for AI inference profiling, infrastructure efficiency, and provider-level optimization analysis.
- Build Python-based analytical workflows, optimization pipelines, dashboards, and infrastructure-aware evaluation systems.
- Authored and contributed to white papers, technical reports, and research initiatives focused on sustainable AI, scalable AI systems, and deployment optimization.
- Work across AI systems topics including model optimization, LLM-oriented workflows, diagnostic tooling, and energy-aware AI infrastructure.

Research Intern - *Cambridge Centre for Alternative Finance (CCAF), University of Cambridge*

Remote | 07/2024 - 06/2025

- Supported research on AI framework development, energy efficiency, model optimization, and AI infrastructure analysis.
- Assisted with fine-tuning, performance analysis, software environment setup, and technical research documentation.
- Contributed to early-stage work on AI energy diagnostics, optimization strategies, and benchmarking concepts.

Research Assistant - *Hamburg University of Technology (TUHH)*

Hamburg, Germany | 04/2024 - 09/2024

- Developed embedded AI and hyperspectral anomaly detection pipelines using deep learning and transfer learning approaches.
- Optimized NVIDIA Jetson Nano, TX2, and AGX Xavier platforms for edge inference, benchmarking, and deployment experimentation.
- Worked with Python, CUDA, C++, MATLAB, Linux, and GitHub for simulation, profiling, and AI deployment workflows.

Research Assistant - *Hanbat National University*

Remote, Daejeon, South Korea | 06/2019 - 08/2021

- Conducted research on communication systems, IoT design, and cognitive network modeling.
- Supported system architecture design, simulation, documentation, and academic writing using MATLAB, Python, LaTeX, and Excel.
- Contributed to peer-reviewed publications in IEEE, Springer, and Wiley venues.

Journal Reviewer - *Science Publishing Group*

Remote, Rockefeller Plaza, New York, | NY 10020 09/2020 - 09/2021

- Reviewed 12+ journal articles and 3 book chapters for Advances in Wireless Communications and Networks (AWCN).

EDUCATION

M.Sc. Microelectronics and Microsystems - *Hamburg University of Technology (TUHH)*

Hamburg, Germany | 10/2021 – 04/2026

- **Specialization:** Embedded Systems and Microsystem Technology.
- **Relevant focus:** GPU architecture, energy efficiency in embedded systems, communication networks, and sustainable technology management.
- **Thesis:** *Hyperspectral Anomaly Detection on Edge Platforms: Transfer-Learning Enhanced Autoencoders with Comparative Multi-Metric Analysis.*

B.Sc. Electrical and Electronic Engineering - *BRAC University*

Dhaka, Bangladesh | 12/2014 - 05/2019

- **Specialization:** Communication Engineering and Computer Engineering.
- **Thesis:** *Performance Analysis of Cognitive Unmanned Aerial Vehicle.* - [Link](#) (Utilizing signal modelling, spectral efficiency analysis, and system simulation)

TECHNICAL SKILLS

Programming: Python, C, C++, MATLAB, R, SQL, Excel VBA

AI / ML / Data: Machine Learning, Deep Learning, Transfer Learning, Model Fine-Tuning, Prompt Engineering, Computer Vision, NLP, LLM Design, RAG, Data Analysis, Model Optimization, ONNX Runtime, Inference Optimization, Pareto Optimization, MLOps Concepts, AI Infrastructure Analysis, Performance Benchmarking

Embedded / Hardware: Embedded Systems, IoT System Design, Hardware Setup and Simulation, Raspberry Pi, Arduino UNO, ATmega32

GPU / Edge Systems: NVIDIA CUDA, GPU Applications, GPU Architecture, NVIDIA Jetson Nano, Jetson TX2, AGX Xavier

Tools / Platforms: Linux, Windows, Git, GitHub, Visual Studio, Docker, FastAPI, Streamlit, ONNX Runtime, GitHub Actions, MATLAB Simulink, Proteus, KiCad, COMSOL, Autodesk Fusion 360, PSpice

Documentation / Research: Scientific Writing, Technical Writing, Academic Publication, Digital Research, Research Methodology, LaTeX Typesetting, Proprietary Data Handling

Office / Productivity: MS Word, Excel, PowerPoint, Access, Publisher, Visio

SELECTED PROJECTS ([GitHub](#))

- *AI Inference Optimization & Deployment Decision Engine:* Built a multi-objective AI deployment optimization platform using Streamlit, ONNX Runtime, Pareto optimization, autoscaling simulation, and carbon-aware analytics. [Link](#)
- *AI Energy Efficiency Index:* Developed a framework for benchmarking AI inference infrastructure efficiency, energy consumption, and deployment optimization. [Link](#)
- *Signal Processing & Intelligent Analytics System:* Built a modular signal analysis and visualization platform with analytical dashboards, intelligent monitoring workflows, and data-driven evaluation pipelines. [Link](#)
- *Edge AI Hyperspectral Anomaly Detection:* Designed & developed transfer-learning autoencoder pipelines on NVIDIA Jetson devices with energy-aware benchmarking and optimization analysis.

PUBLICATIONS

- *Cognitive Unmanned Aerial Vehicle-aided Human Bond Communication System: Modeling and Performance Analysis.* IEEE Access, 2022 - [Link](#)
- *An optimization based approach to enhance the throughput and energy efficiency for cognitive unmanned aerial vehicle networks.* Wireless Networks, Springer, 2021 - [Link](#)
- *Cognitive IoT-Based Health Monitoring Scheme Using Non-Orthogonal Multiple Access.* Wiley Book Chapter, 2022 - [Link](#)

AWARDS & LANGUAGES

Languages: Bengali (Native), English (C2), German (A2).

Awards: Vice Chancellor's List Award, BRAC University; Dean's List Award, BRAC University.